University of Toronto

AER1515 Perception for Robotics: Assignment 3

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Contents

1 Answers

1.1 Part 1: Depth Estimation

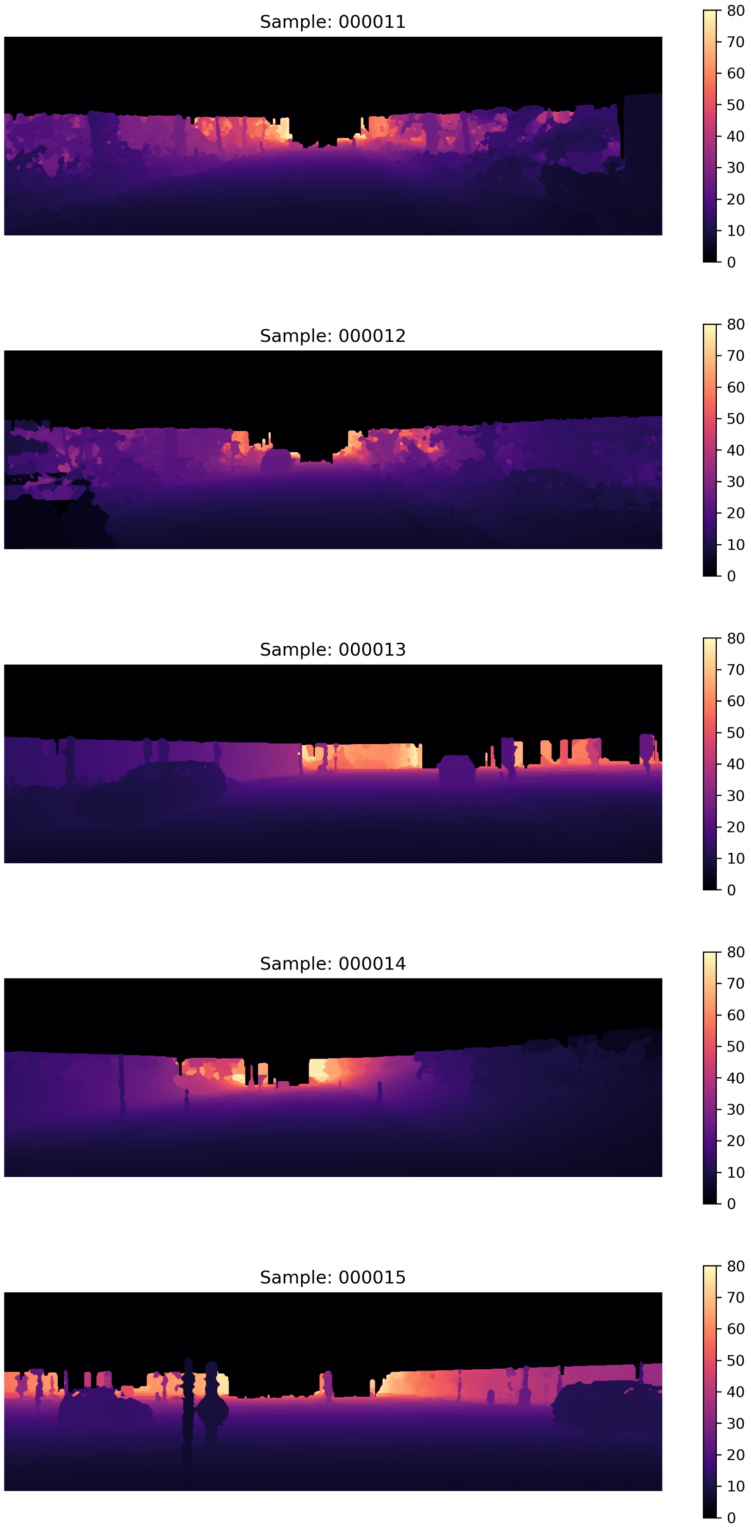


Figure 1: Estimateded depth using dense disparity maps. The color bar represents the depth in meters.

Observations on the Depth Image Quality The produced depth images show reasonable consistency across the scene, with smooth gradients along the road and clear separation between near and far objects. Depth accuracy is generally higher for textured and well-lit surfaces, such as the road and nearby vehicles, where disparity estimation remains stable and consistent. In contrast, reflective or low-texture areas, tree canopies, and occlusion boundaries often exhibit noisy or discontinuous depth values due to unreliable or missing disparity information.

1.2 Part 2: 2D Bounding Box Detection

Figure 2: Estimated 2D bounding box using the YOLOv3 object detector with `confidence_threshold:0.51` and `NMS_threshold:0.6`

1.3 Part 3: Instance Segmentation

To perform the instance segmentation task, I implemented two following approaches:

1. Basic Method
2. YOLOv11x-seg model

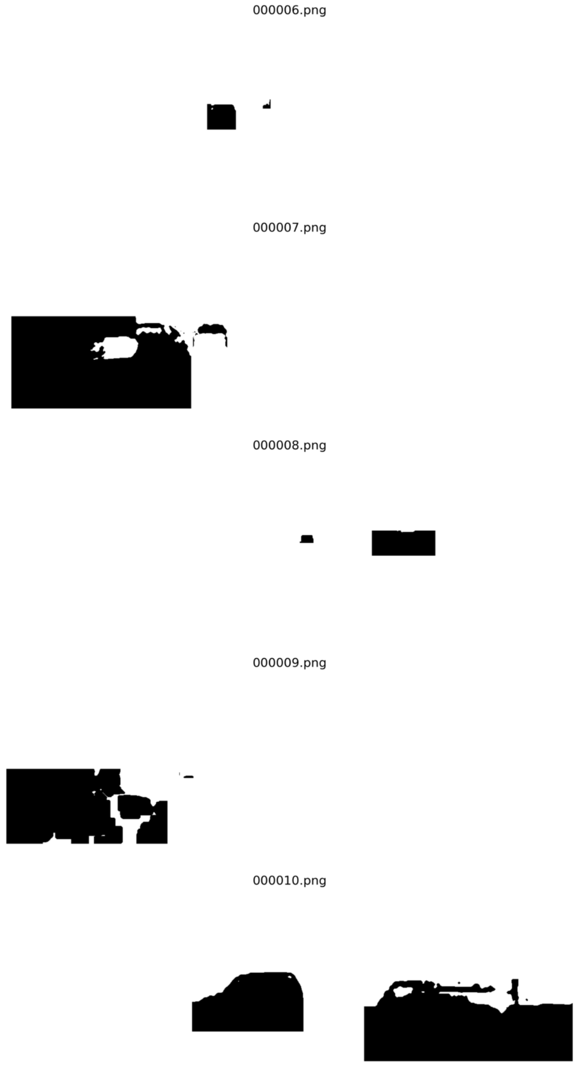
1.3.1 Basic Method

A typical non-learning method to perform instance segmentation is to:

1. Calculate the average depth within the given bounding boxes
2. Set a distance threshold from the computed average depth and assign fitting pixels as the object's segmentation mask

The result are displayed below. Note that for the python scripts to generate segmentation masks using the basic methods are limited to the train datasets, as the performance are deemed to be worse than learning methods, so I utilized the ground truth depth map and bounding boxes provided with the data.

(a) 2D bounding boxes for train images.



(b) Basic instance segmentation method with `distance_threshold:5.8`

Figure 3

From Fig. 3, we can see that the segmentation mask genreated by the basic method aren't very accurate. Therefore, it's time to utilize the power of deep learning methods.

1.3.2 YOLOv11x-seg Instance Segmentation Model

YOLOv11x-seg is an advanced instance segmentation model developed by Ultralytics [1], based on the YOLO (You Only Look Once) architecture. It performs real-time object detection and segmentation in a single forward pass through the network. The model consists of a convolutional backbone for feature extraction, a neck with a feature pyramid network (FPN) for multi-scale feature fusion, and a detection head that predicts bounding boxes, class probabilities, and pixel-level masks for each detected object. Trained on large datasets like COCO, it uses anchor-free predictions and advanced loss functions to achieve high accuracy and speed. The following Fig. 4 shows the 2D bounding box and instance segmentation masks using this architecture.

(a) 2D bounding boxes detected by the YOLOv11x model.

(b) Instance segmentation masks for cars; generated by the YOLOv11x model.

Figure 4

References

[1] G. Jocher and J. Qiu, "Ultralytics yolo11," 2024.